# PROJECT DOCUMENTATION

# MICROGOAL

# 1. Introduction

**Project Title:** MICROGOAL

**Team ID:** SWTID-2026-4792

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# 2. Project Overview

**Purpose:** MicroGoal is a productivity and goal■tracking platform designed to help users break large goals into smaller achievable tasks called micro goals. The system encourages daily progress, motivation, and effective time management.

**Features:**

• Goal creation and micro■task breakdown

• Daily progress tracking

• Reminders and notifications for tasks

• User dashboard for monitoring progress

• Admin panel for managing users and content

• Secure login and profile management

# 3. Architecture

Frontend: React.js with Bootstrap for responsive user interface.

Backend: Node.js with Express.js for server-side logic.

Database: MongoDB with Mongoose for storing user goals and progress.

# 4. Tools and Technologies

• React.js

• Node.js

• Express.js

• MongoDB

• Mongoose

• Git

• Visual Studio Code

## 5. Folder Structure

```
MicroGoal/
|-- client/ (React frontend)
|---- components/
|---- pages/
|-- server/ (Node.js backend)
|---- routes/
|---- models/
|---- controllers/
```

## 6. Running the Application

• Clone the repository from GitHub.

• Install frontend dependencies using npm install in the client folder.

• Install backend dependencies using npm install in the server folder.

• Run frontend using npm start.

• Run backend server using node server.js or npm start.

## 7. API Endpoints

• /api/goals/create – Create a new goal

• /api/goals/:id – Get goal details

• /api/tasks/add – Add micro tasks

• /api/tasks/update – Update progress

• /api/users/login – User authentication

## 8. Authentication

The system uses JWT-based authentication to secure user login and protect private routes. Middleware ensures that only authenticated users can access their goal data.

## 9. User Interface

• Landing Page

• User Dashboard

• Goal Creation Page

• Task Progress Tracker

• Admin Panel

## 10. Testing

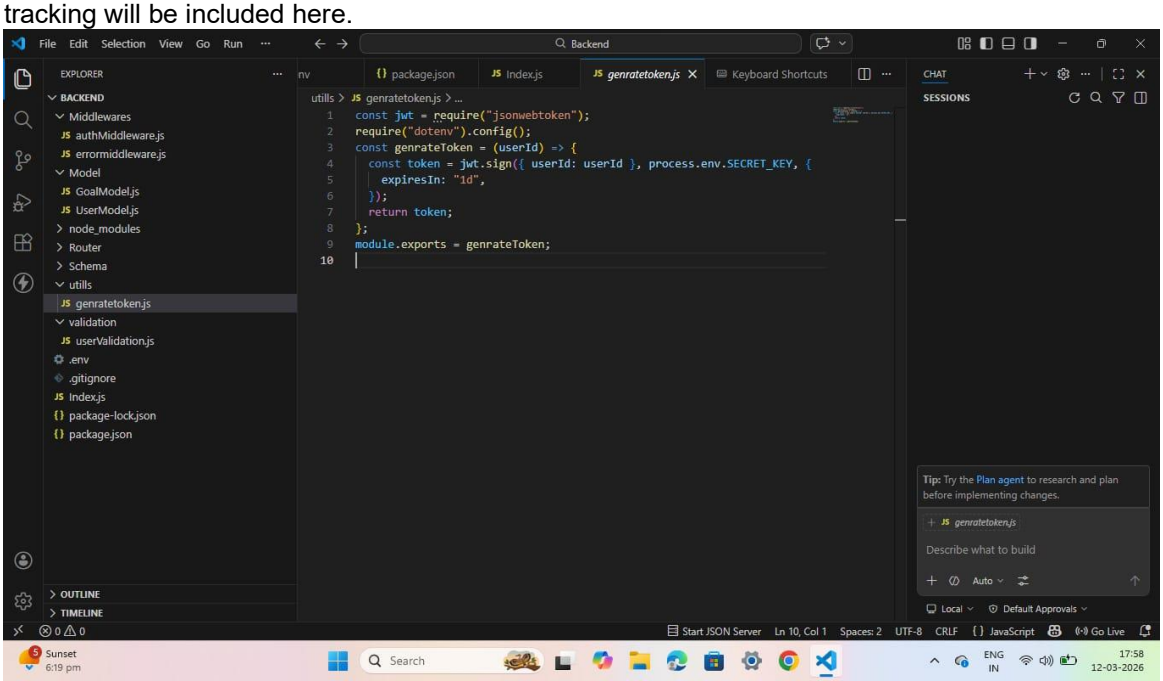
• Manual testing during development stages

• API testing using Postman

• Browser testing using Chrome Developer Tools

## 11. Screenshots or Demo

Screenshots of the application interface such as dashboard, goal creation page, and progress tracking will be included here.



File Edit Selection View Go Run ... Backend

EXPLORER

BACKEND
- Middlewares
  - authMiddleware.js
  - errormiddleware.js
- Model
- node_modules
- Router
- Schema
- utills
  - genratetoken.js
- validation
  - userValidation.js
- .env
- .gitignore
- Index.js
- package-lock.json
- package.json

OUTLINE

TIMELINE

.env | package.json | Index.js | authMiddleware.js | Keyboard Shortcuts

Middlewares > authMiddleware.js > ...

```
1  const jwt = require("jsonwebtoken");
2  require("dotenv").config();
3
4  const authmiddleware = (req, res, next) => {
5    try {
6      const header = req.headers.authorization;
7
8      if (!header) {
9        return res.status(401).json({ message: "No token" });
10     }
11
12     const token = header.split(" ")[1];
13
14     const decoded = jwt.verify(token, process.env.SECRET_KEY);
15
16     req.user = decoded.userId;
17
18     next();
19   } catch (error) {
20     return res.status(401).json({ message: "Invalid token" });
21   }
22 };
23 module.exports = authmiddleware;
24
```

CHAT

SESSIONS

Tip: Try the Plan agent to research and plan before implementing changes.

authMiddleware.js

Describe what to build

Auto

Local | Default Approvals

0 0 | Start JSON Server | Ln 1, Col 1 | Spaces: 2 | UTF-8 | CRLF | JavaScript | Go Live

Sunset 6:19 pm | Search | ENG IN | 18:11 12-03-2026

## 12. Known Issues

Some minor performance delays may occur when loading a large number of goals. Future optimization will improve loading speed and database queries.

## 13. Future Enhancements

- Mobile application support
- AI-based goal suggestions
- Gamification with rewards and achievements
- Calendar integration
- Team goal collaboration features